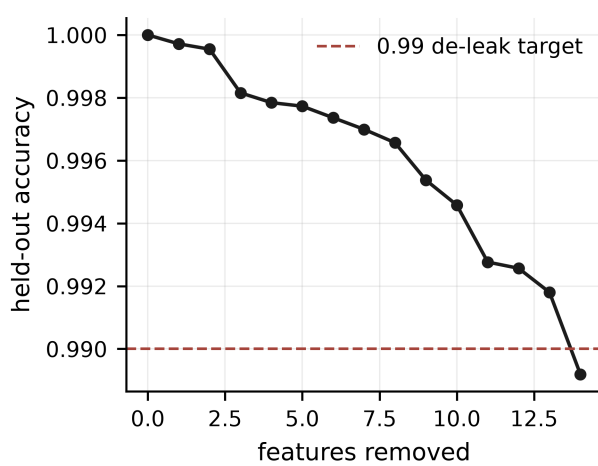
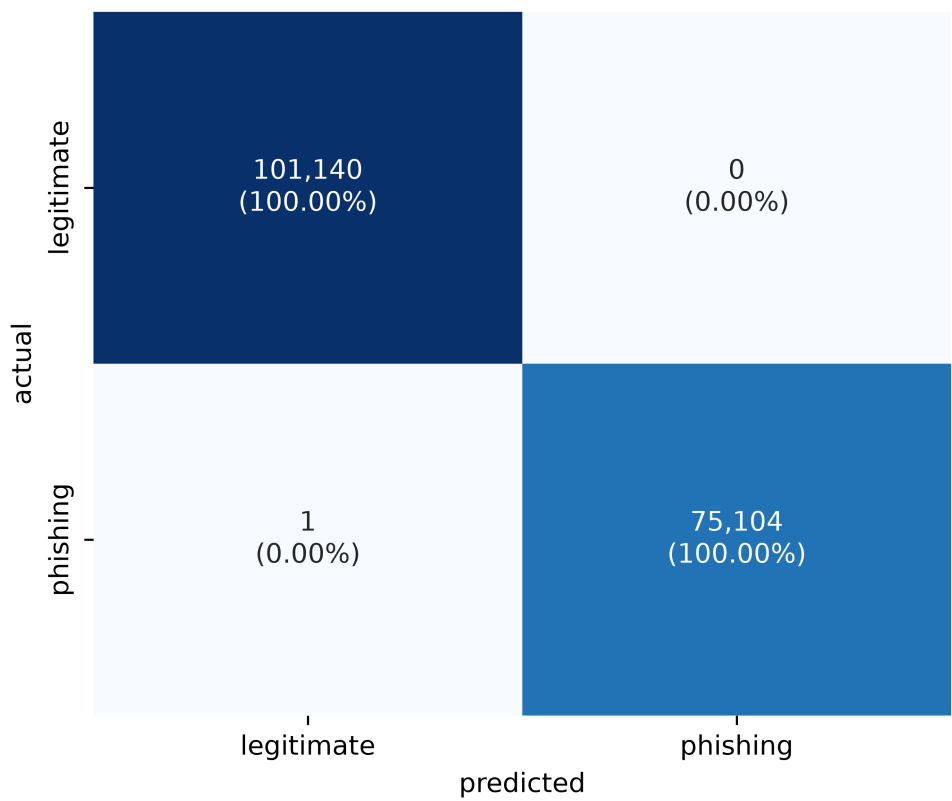


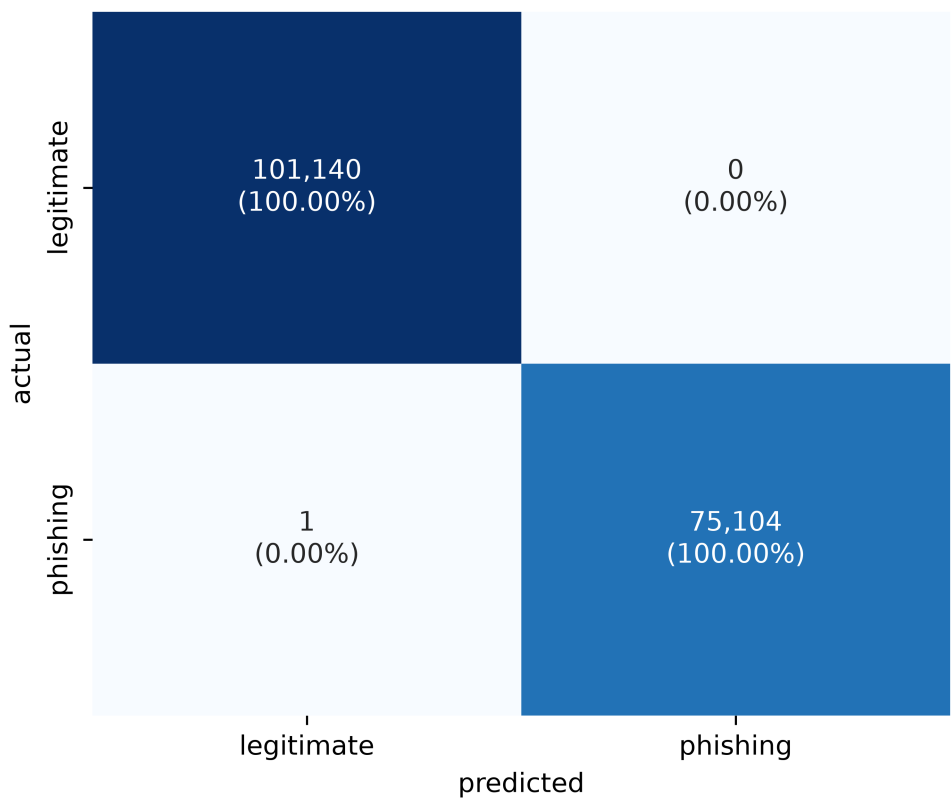
Progressive removal on PhiUSIL: held-out accuracy against the number of features removed.



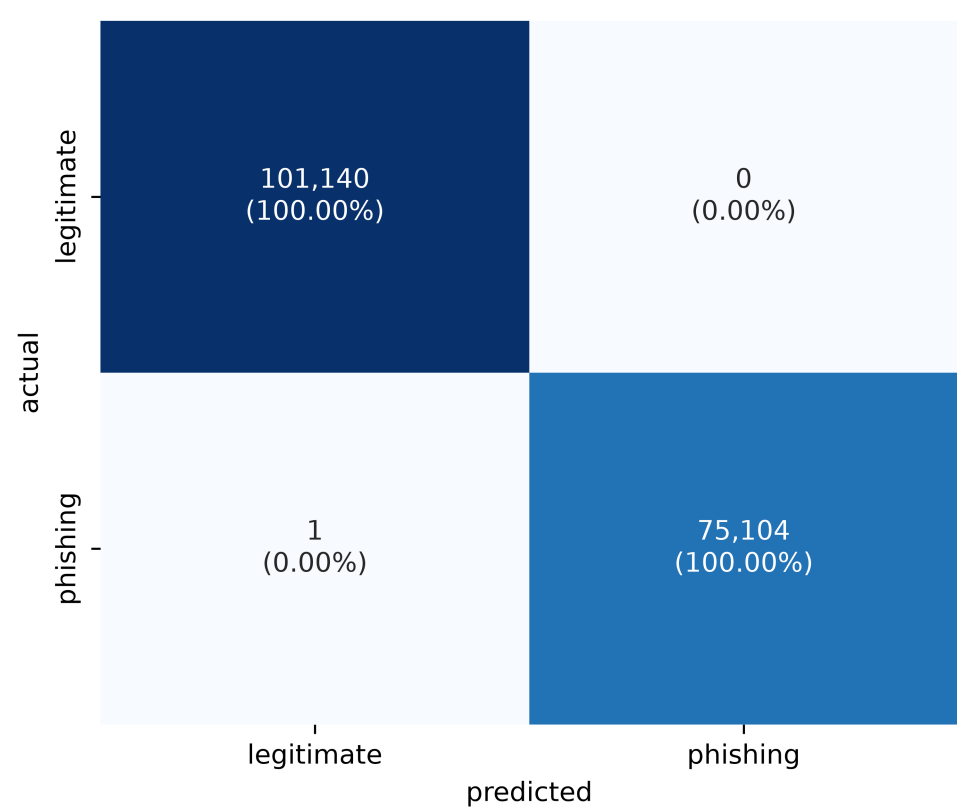
Confusion matrix, summed over five seeds: PhiUSIIL - Random Forest.



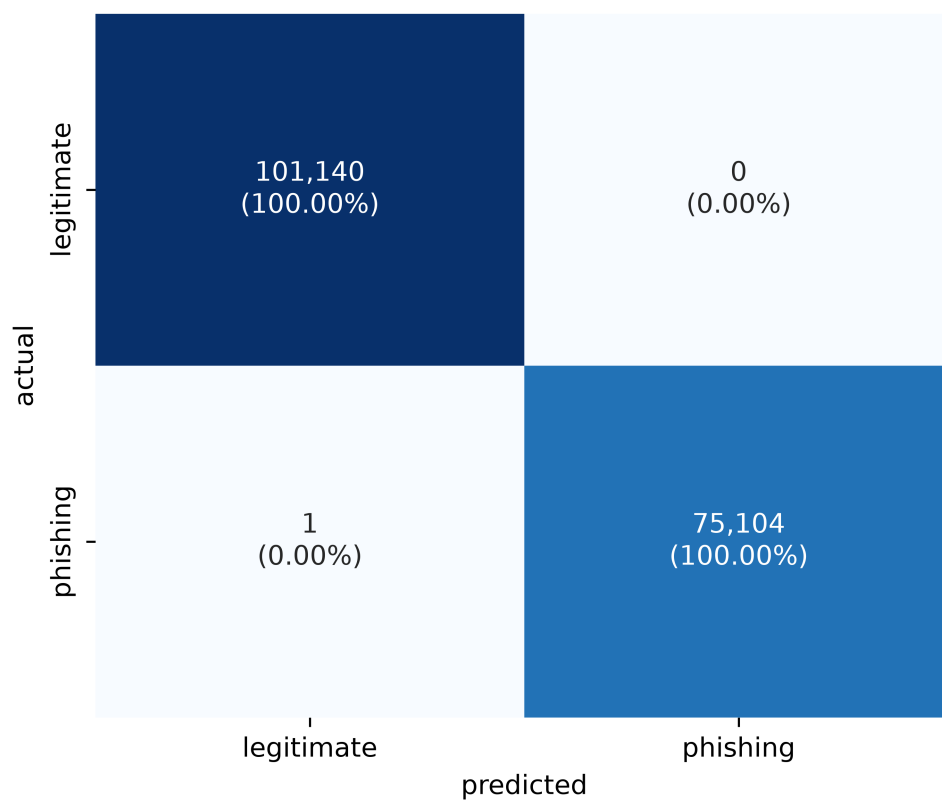
Confusion matrix, summed over five seeds: PhiUSIIL - XGBoost.



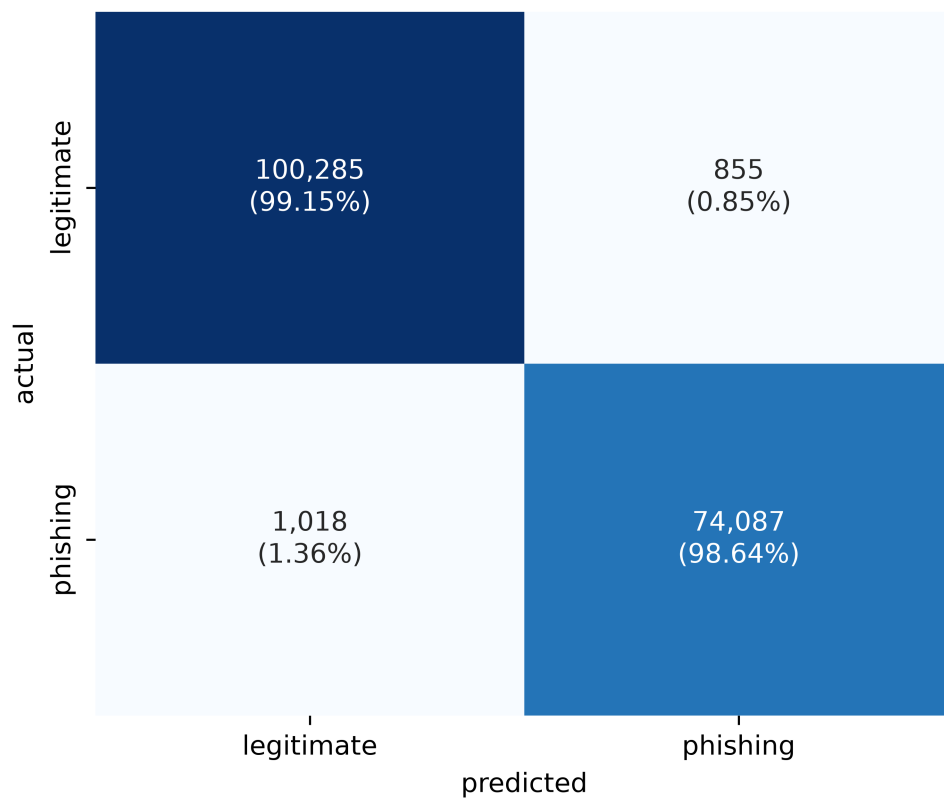
Confusion matrix, summed over five seeds: PhiUSIIL - LightGBM.



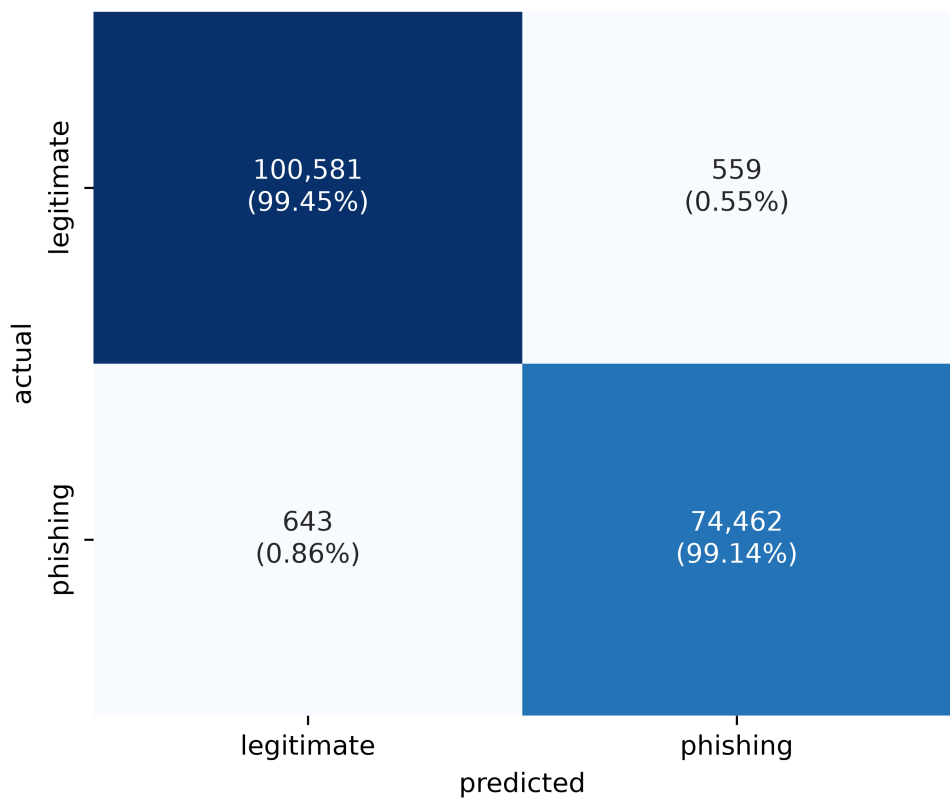
Confusion matrix, summed over five seeds: PhiUSIIL - Soft-voting ensemble.



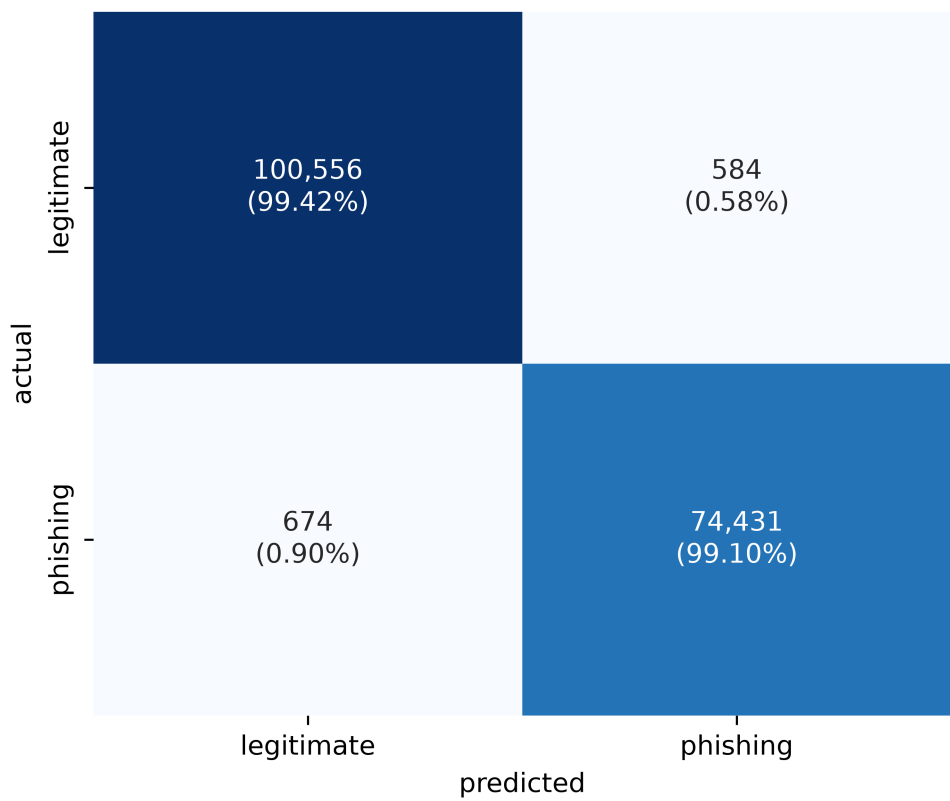
Confusion matrix, summed over five seeds: PhiUSIL (leak-controlled) - Random Forest.



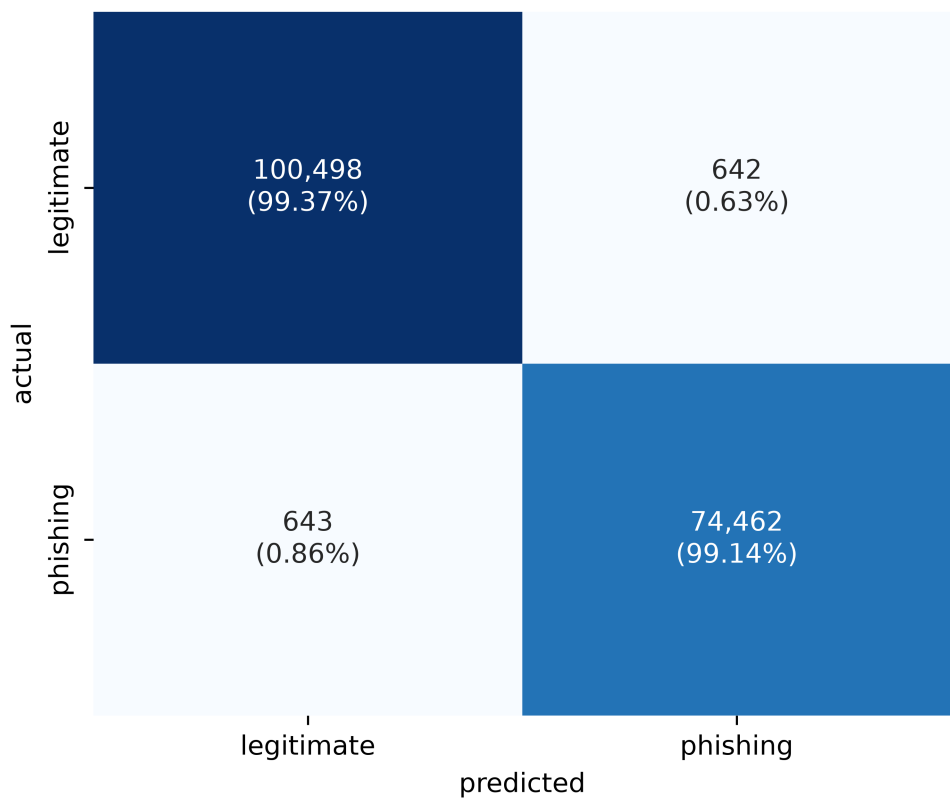
Confusion matrix, summed over five seeds: PhiUSIIL (leak-controlled) - XGBoost.



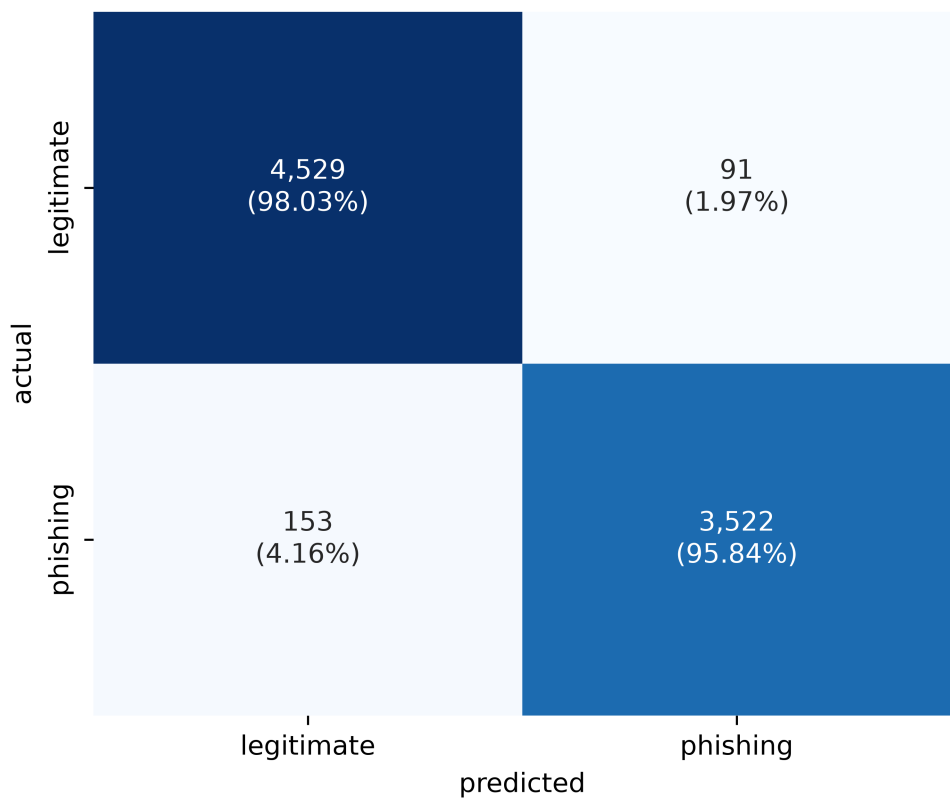
Confusion matrix, summed over five seeds: PhiUSIIL (leak-controlled) - LightGBM.



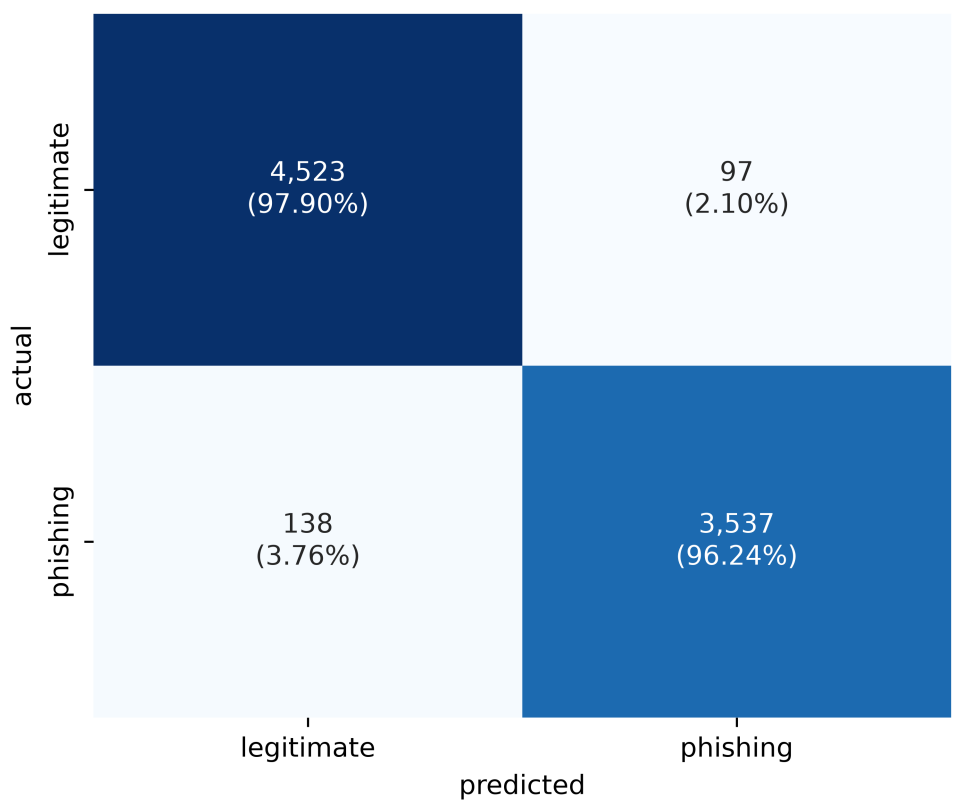
Confusion matrix, summed over five seeds: PhiUSIL (leak-controlled) - Soft-voting ensemble.



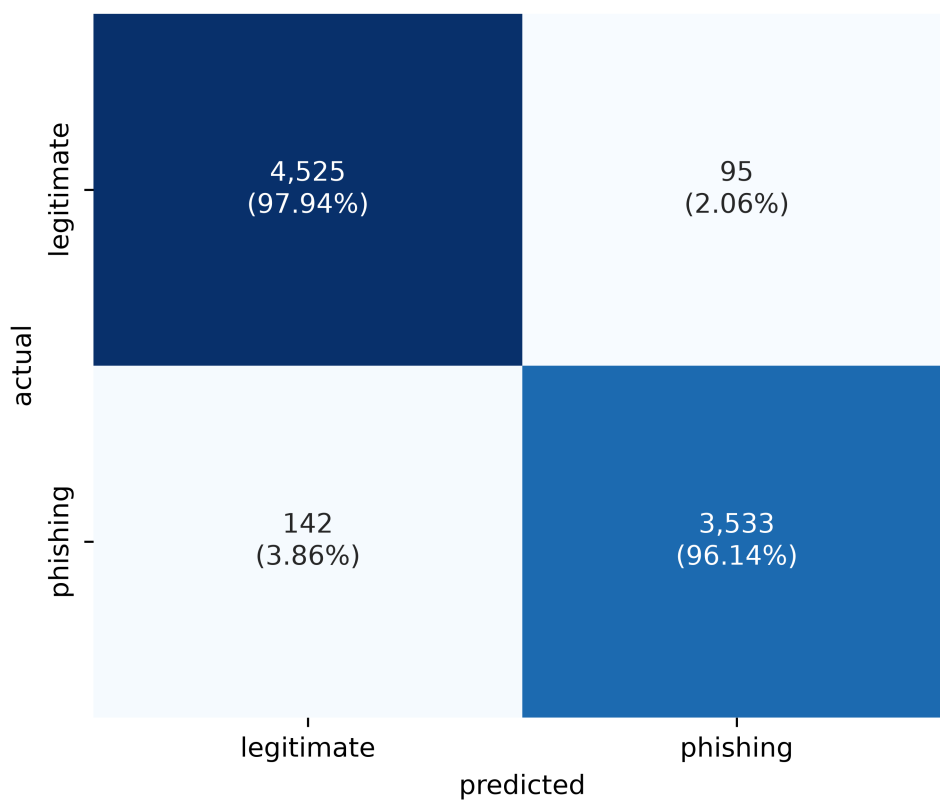
Confusion matrix, summed over five seeds: UCI Phishing Websites - Random Forest.



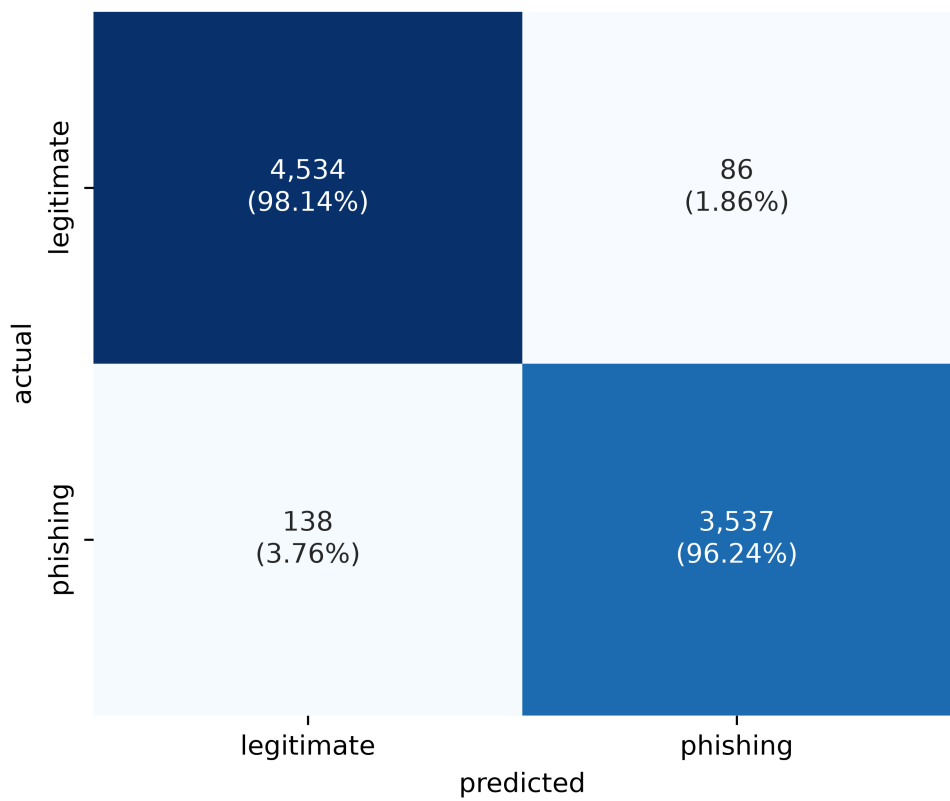
Confusion matrix, summed over five seeds: UCI Phishing Websites - XGBoost.



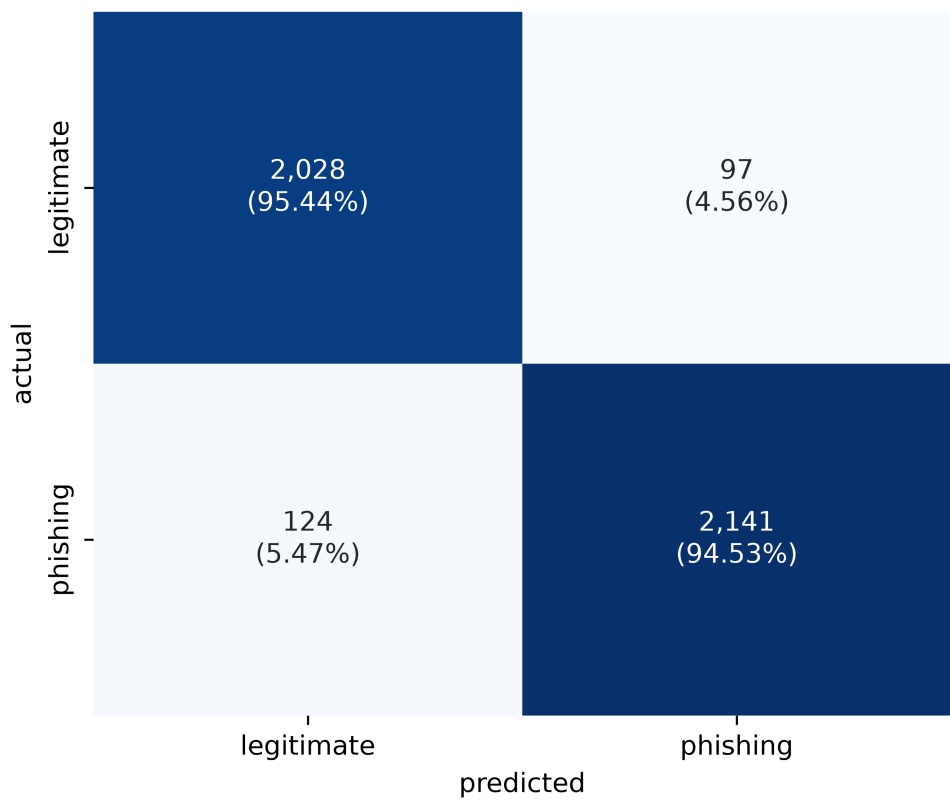
Confusion matrix, summed over five seeds: UCI Phishing Websites - LightGBM.



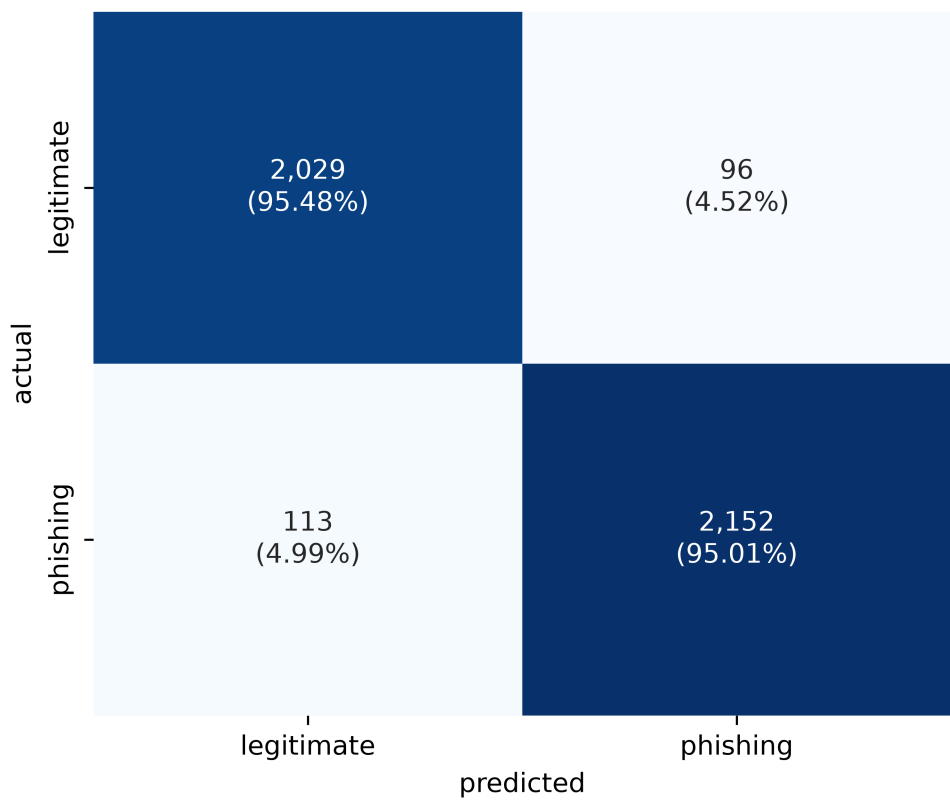
Confusion matrix, summed over five seeds: UCI Phishing Websites - Soft-voting ensemble.



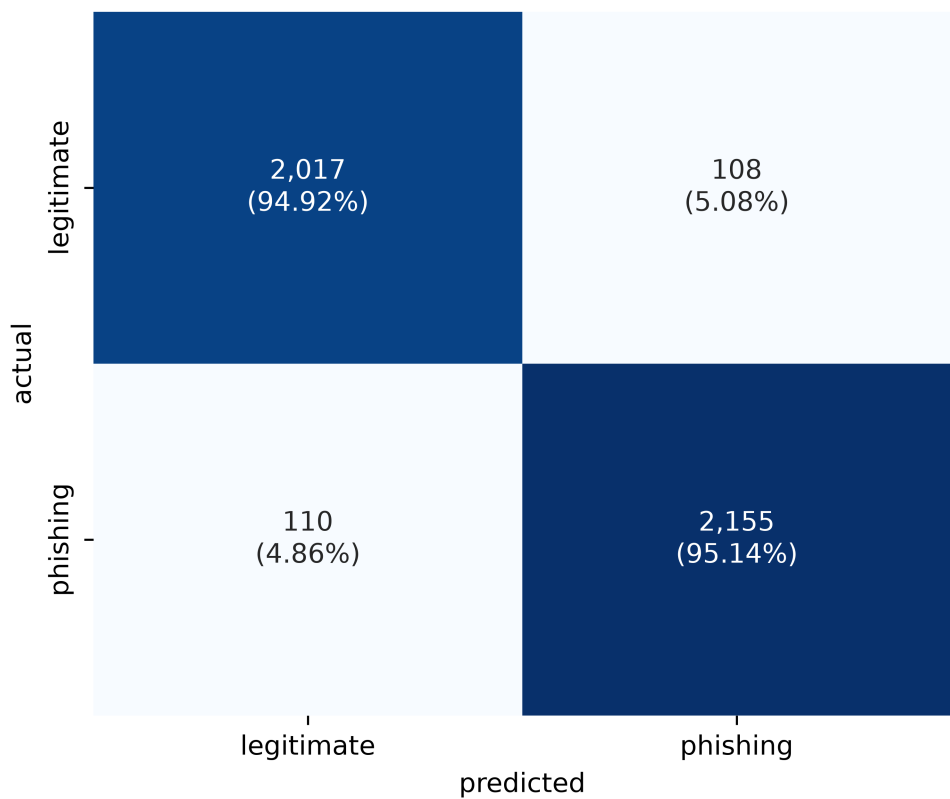
Confusion matrix, summed over five seeds: UCI Phishing Websites (de-duplicated) - Random Forest.



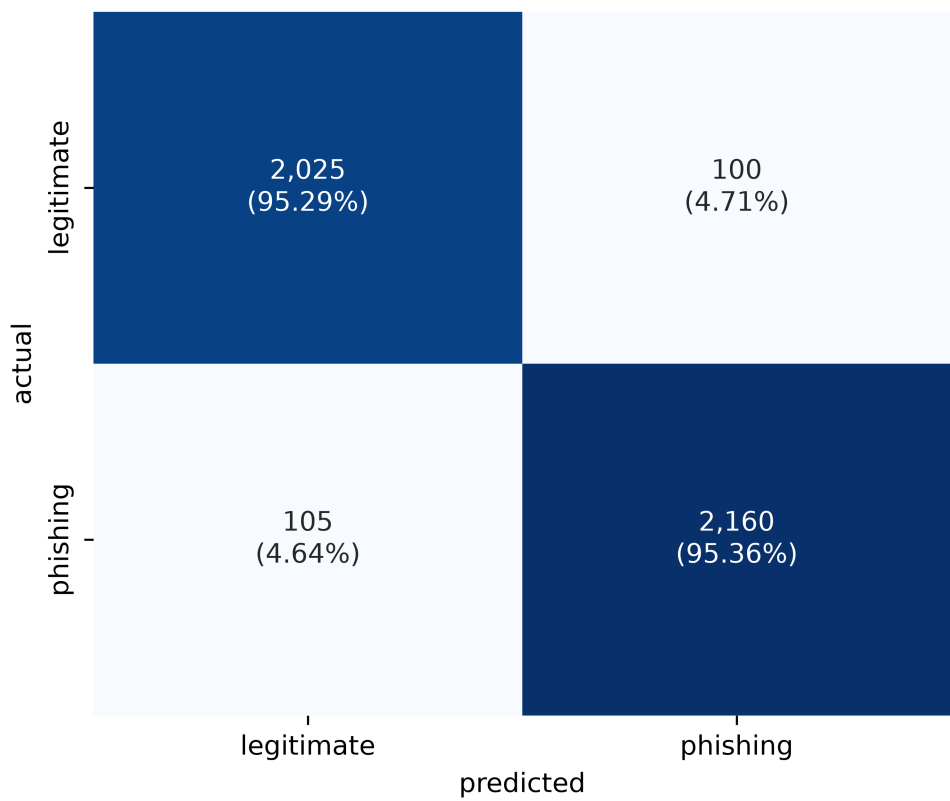
Confusion matrix, summed over five seeds: UCI Phishing Websites (de-duplicated) - XGBoost.



Confusion matrix, summed over five seeds: UCI Phishing Websites (de-duplicated) - LightGBM.



Confusion matrix, summed over five seeds: UCI Phishing Websites (de-duplicated) - Soft-voting ensemble.



SHAP sample-size convergence check: Kendall's tau between attribution rankings computed at successive sample sizes.

dataset	size_a	size_b	effective_a	effective_b	kendall_tau	n_available	degenerate
PhiUSIIL	5000	10000	5000	10000	0.9984	35249	no
PhiUSIIL	10000	20000	10000	20000	0.9967	35249	no
UCI Phishing Websites	219	439	219	439	0.9310	878	no
UCI Phishing Websites	439	878	439	878	0.9770	878	no
UCI Phishing Websites	878	5000	878	878	1.0000	878	YES - see note
UCI Phishing Websites	5000	10000	878	878	1.0000	878	YES - see note
UCI Phishing Websites	10000	20000	878	878	1.0000	878	YES - see note

Note on the rows marked degenerate:

UCI Phishing Websites has a held-out pool of 878 instances. The 878/5000, 5000/10000 and 10000/20000 comparisons therefore draw the same sample at both nominal sizes, so their tau = 1.0000 holds by construction and is not evidence of convergence. The informative UCI rows are the adaptive ladder (219/439 and 439/878). PhiUSIIL's pool is large enough for the requested ladder to run in full.